

Windows Host Investigation – Incident Investigation Report

Platform: TryHackMe

Investigation Type: Windows Host / Digital Forensics Investigation

Assessment: Post-compromise investigation

1. Executive Summary

The Windows system investigated in this exercise showed signs of suspicious activity and potential compromise. Investigation of user accounts, login activity, temporary files, and unusual files revealed evidence that an unauthorized user may have interacted with the system and attempted to hide or store malicious activity. An unusual temporary directory and a suspicious file associated with Windows password information were identified during the investigation. The system should therefore be treated as compromised until the suspicious activity has been fully investigated, the affected accounts have been secured, and the host has either been securely cleaned or rebuilt.

2. Scope & Data Sources

The investigation focused on a Windows host provided as part of the TryHackMe Investigating Windows exercise.

The following information and Windows artefacts were examined:

Data Source / Artefact	Purpose	What It Could Tell Us
Windows command prompt	Host investigation and enumeration	System configuration and local account information
<code>systeminfo</code>	Operating system identification	Windows version and system information
<code>net user</code>	Account enumeration	Users present on the system and account-related information
<code>net user <username></code>	Individual account investigation	Information about specific users and their activity

User logon information	Authentication investigation	Evidence of users accessing the system
Temporary directories	File-system investigation	Unusual files and directories that may have been created during an intrusion
Files located under the suspicious TMP directory	Malware/artifact investigation	Potential tools, payloads, or stolen data
RDP connection	Initial access to the lab system	Allowed the investigation to be conducted remotely

The investigation began by connecting to the Windows host through Remote Desktop. The notes document the use of [xfreerdp](#) to establish the connection and subsequently perform host enumeration.

Limitations

The available evidence was primarily focused on basic Windows host and file-system investigation. The supplied notes do not document a complete examination of:

- Windows Security Event Logs
- PowerShell operational logs
- Sysmon telemetry
- Windows Defender logs
- Prefetch files
- Amcache/Shimcache
- Registry hives
- Scheduled Tasks
- Services
- Browser history
- Network traffic
- File hashes

Consequently, conclusions about the exact initial-access technique, complete attacker timeline, and persistence mechanism should be treated cautiously.

3. Timeline of Events

The available investigation notes do not provide exact timestamps for every activity. Therefore, the timeline below separates confirmed investigative observations from conclusions that cannot currently be timestamped.

Order	Time	Event	Significance
1	Initial investigation	Analyst established an RDP connection to the Windows host	Provided remote access for the investigation
2	Initial investigation	<code>systeminfo</code> was used to identify the Windows version	Established the operating-system context
3	Initial investigation	Local users were enumerated using <code>net user</code>	Identified accounts that could be investigated
4	Investigation	Administrator-level users were identified	Important because privileged accounts could make significant system changes
5	Investigation	John's login activity was examined	Provided evidence for user activity
6	Investigation	Jenny's last logon information was examined	Established additional user activity
7	File-system investigation	An unusual TMP directory was identified	Potential indication of attacker activity
8	File investigation	Files were discovered underneath the suspicious TMP location	Required further examination because temporary directories can be abused to stage tools or data
9	File investigation	A suspicious file was identified that appeared related to Windows passwords	Potentially significant because it may indicate credential-access activity

The notes specifically identify an unusually located TMP directory and suspicious files underneath it. The investigation also identified a file whose name/content appeared to point toward Windows passwords.

Timeline Limitation

Exact timestamps for these events were not preserved in the supplied notes. A production investigation should obtain timestamps from Windows event logs, file metadata, registry artefacts, and other forensic sources before establishing a definitive incident timeline.

4. Investigation & Analysis

4.1 Remote Access

The investigation began by connecting to the Windows system through Remote Desktop.

The documented command was:

```
xfreerdp /v:10.67.149.208 /u:Administrator /p:'letmein123!' /cert:ignore
```

This provided access to the Windows system using the Administrator account.

Security significance:

Use of an Administrator account for remote access is significant because compromise of that account would give an attacker extensive control over the host.

The credentials shown above should be considered TryHackMe lab credentials, not production credentials.

4.2 Operating System Identification

`systeminfo` was used to determine the Windows operating-system version and associated information.

The notes specifically identify:

```
systeminfo  
systeminfo | findstr /B /C:"OS Name"
```

as methods for determining the Windows version and year.

This step established the baseline environment in which the remaining investigation was performed.

4.3 User Account Investigation

The investigation then focused on identifying accounts present on the system.

The `net user` command was used to enumerate users, while:

```
net user <username>
```

could be used to obtain information about an individual account.

The investigation identified multiple users, including John and Jenny, and examined their login information.

Administrator-level accounts were also examined.

Assessment

User-account investigation is important because an attacker may:

1. Compromise an existing account.
2. Create a new account.
3. Abuse an account with excessive privileges.
4. Use an existing privileged account to avoid creating a new account.

The evidence available in the notes establishes that multiple user accounts and their logon information were investigated. However, it does not by itself prove which account was initially compromised.

4.4 Suspicious Temporary Directory

A particularly important finding was the discovery of a TMP directory in an unusual location.

The investigation notes state:

"The TMP was created at an odd location"

and subsequently identify files located underneath the TMP directory.

Assessment

Temporary directories are not inherently malicious. Legitimate Windows applications regularly create temporary files.

However, an unusual temporary directory containing suspicious files becomes more significant when considered alongside:

- Suspicious user activity
- Potential credential-related files
- Unusual file locations
- Possible attempts to conceal attacker tooling

Therefore, the directory should be treated as a potential staging or execution location until its contents and creation history are fully understood.

4.5 Suspicious Password-Related File

The investigation identified a file under the suspicious temporary directory that appeared to reference Windows passwords.

The notes indicate that the filename/content pointed toward something resembling a tool associated with Windows passwords.

Assessment

This is one of the most significant findings in the investigation.

A password-related utility located in an unusual temporary directory could indicate that someone was attempting to:

- Obtain credentials
- Extract password information
- Prepare for credential-based lateral movement
- Stage a credential-access tool

The available evidence does not, by itself, establish that credentials were successfully extracted or transmitted off the machine.

What Was Ruled Out

The investigation did not find sufficient evidence in the supplied notes to prove:

- Successful credential exfiltration
- Lateral movement to another system
- Data destruction
- Ransomware deployment
- Confirmed malware execution
- Persistence through a scheduled task
- Persistence through a Windows service

These should remain investigation questions rather than being presented as confirmed facts.

5. Persistence Mechanisms

Persistence is the ability of an attacker to maintain access after events such as a reboot, logout, or password change.

Based on the supplied investigation notes, no persistence mechanism is conclusively documented.

However, the suspicious files and unusual temporary directory warrant investigation of common Windows persistence locations.

Persistence Mechanism	Investigation Required	Finding
Scheduled Tasks	Review Task Scheduler and task files	Not documented in supplied notes
Windows Services	Review newly created/unusual services	Not documented
Registry Run Keys	Review Run and RunOnce keys	Not documented
Startup Folder	Review user/system startup locations	Not documented
New User Accounts	Compare account list with known users	Accounts were investigated, but persistence was not conclusively established
WMI Event Subscriptions	Review permanent WMI subscriptions	Not documented
PowerShell Profiles	Inspect user profiles	Not documented
Remote Desktop configuration	Review RDP-related configuration and logs	RDP was used during the investigation, but persistence was not established

Persistence Assessment

The absence of a documented persistence mechanism does not prove that persistence did not exist.

A complete forensic investigation should inspect the above locations and correlate any suspicious entries with creation timestamps, file metadata, user activity, and Windows event logs.

6. Indicators of Compromise

The following indicators are based on the evidence available from the investigation notes.

Indicator	Type	Significance	Recommended Action
10.67.149.208	IP address	IP address of the investigated lab host	Treat as lab infrastructure; do not block in production
Administrator	Account	Privileged account used for RDP access	Review authentication activity
John	User account	User identified during investigation	Review logon history
Jenny	User account	User identified during investigation	Review last-logon activity
Unusual TMP directory	Directory	Suspicious temporary location	Investigate creation and contents
Files under TMP directory	Files	Potential attacker tooling/staging	Preserve and hash
Password-related suspicious file	File	Potential credential-access tooling	Isolate and perform malware/forensic analysis
RDP access	Technique	Remote access mechanism	Review Windows Security logs for RDP authentication
Administrator privileges	Privilege	Could provide extensive system control	Audit privileged account use

Additional IoCs That Should Be Collected

A production-quality investigation should additionally record:

- Full file paths
- SHA-256 hashes

- File creation/modification/access times
- File owners
- File permissions
- Process names
- Process IDs
- Parent processes
- Source IP addresses
- Destination IP addresses
- Scheduled task names
- Service names
- Registry keys
- PowerShell commands
- Windows Event IDs
- User Security Identifiers (SIDs)

These values are not available in the supplied notes and therefore have not been invented.

7. Containment & Recommendations

7.1 Actions for the Next Hour

The priority should be to prevent further unauthorized access while preserving evidence.

1. Isolate the Windows host

Disconnect the affected host from unnecessary network access.

Do not immediately delete suspicious files because doing so may destroy evidence needed to determine what happened.

2. Protect privileged accounts

Immediately reset credentials for accounts suspected of being exposed, particularly privileged accounts.

Where possible:

- Disable unused accounts.
- Remove unnecessary administrative privileges.
- Require strong passwords.
- Enable multi-factor authentication for remote access where supported.

3. Preserve suspicious files

The suspicious TMP directory and its contents should be preserved for analysis.

Calculate cryptographic hashes such as SHA-256 before making modifications.

4. Review authentication logs

Examine Windows Security logs for:

- Successful logons
- Failed logons
- Remote interactive logons
- Account creation
- Privilege assignment
- RDP authentication

5. Investigate persistence

Immediately review:

- Scheduled Tasks
- Services
- Registry Run/RunOnce keys
- Startup folders
- WMI subscriptions
- New accounts
- PowerShell profiles

6. Review network activity

Determine whether the suspicious system communicated with external or internal systems during the suspected compromise period.

8. Detection Recommendations

A detection should have alerted defenders to the combination of suspicious account activity and credential-access tooling.

One useful detection would correlate:

Privileged account logon + suspicious temporary-file creation + password/credential-access tooling.

For example:

IF
 a privileged account performs a remote logon
AND
 a new executable/script is created in a temporary directory
AND
 the filename or process behavior indicates credential access
THEN
 generate a HIGH severity security alert

This would be significantly stronger than alerting on any one event individually.

Example Detection Scenario

User: Administrator

Logon Type: Remote Interactive

↓

Process created from suspicious TMP directory

↓

Process/file associated with password access

↓

HIGH-SEVERITY ALERT

This type of correlation could have reduced the time required to identify the activity.

9. Recovery Decision – Clean or Rebuild

Given the evidence identified during this investigation, rebuilding the host is the preferred response if the system were a production endpoint.

The reason is the uncertainty surrounding the extent of the compromise. The investigation identified suspicious files in an unusual temporary location and evidence pointing toward password-related activity. If an attacker obtained administrative-level access, defenders cannot reliably assume that deleting the obvious suspicious file would remove every modification made to the system.

An attacker with sufficient privileges could potentially establish persistence through mechanisms that are not immediately visible.

Therefore:

Preferred action:

Preserve the affected system for forensic purposes, collect the required evidence, reset potentially exposed credentials, and rebuild the host from a known-good image.

For a disposable TryHackMe machine, this level of recovery is unnecessary once the exercise is complete. For an enterprise workstation or server, however, rebuilding provides greater confidence that hidden modifications have been removed.

10. Lessons Learned

Several important lessons can be taken from this investigation.

1. Unusual file locations matter

A file does not need to have an obviously malicious name to deserve investigation. An unusual location combined with suspicious content can be a strong indicator.

2. User activity provides important context

Investigating accounts and last-logon information can help establish who interacted with a system and whether account activity requires further investigation. The investigation specifically examined accounts including John and Jenny.

3. Privileged accounts require additional scrutiny

Administrative access significantly increases the potential impact of a compromise.

4. Persistence must always be investigated

Finding the suspicious file is not the end of the investigation. Analysts must determine whether the attacker created a mechanism that would allow them to return after a reboot or credential reset.

5. Evidence preservation is critical

Deleting suspicious files immediately can remove important forensic evidence. Investigation should prioritize preservation, documentation, hashing, and analysis before remediation.

11. Final Assessment

The investigation identified multiple indicators requiring security attention, including unusual user activity, an oddly located temporary directory, suspicious files beneath that directory, and a file apparently associated with Windows password access.

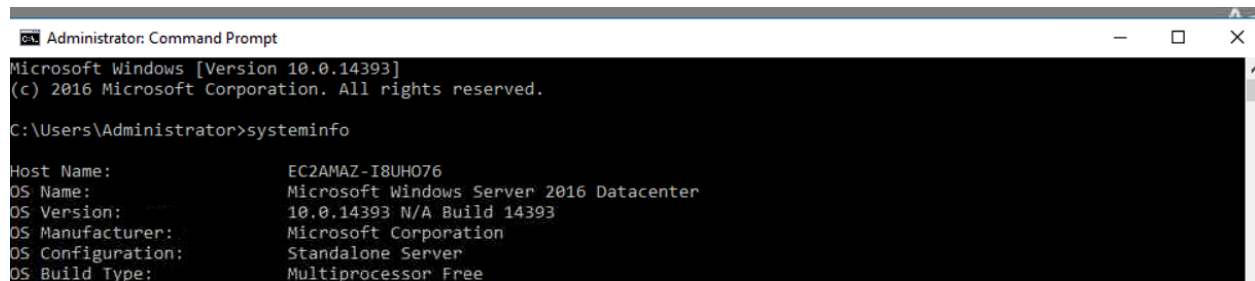
The available evidence supports treating the Windows host as potentially compromised, but the supplied investigation notes do not contain enough evidence to establish the complete attack chain, exact timestamps, successful credential theft, data exfiltration, or a specific persistence mechanism.

The appropriate response is therefore to preserve the evidence, investigate authentication and persistence mechanisms, secure potentially exposed accounts, and—if this were a real production host—rebuild the system from a trusted image after forensic collection.

Screenshots and filters/commands used.

xfreerdp /v:10.67.149.208 /u:Administrator /p:'letmein123!' /cert:ignore - RDP connection

Systeminfo or systeminfo | findstr /B /C:"OS Name" for a more specific response - on cmd to find the windows version and the year

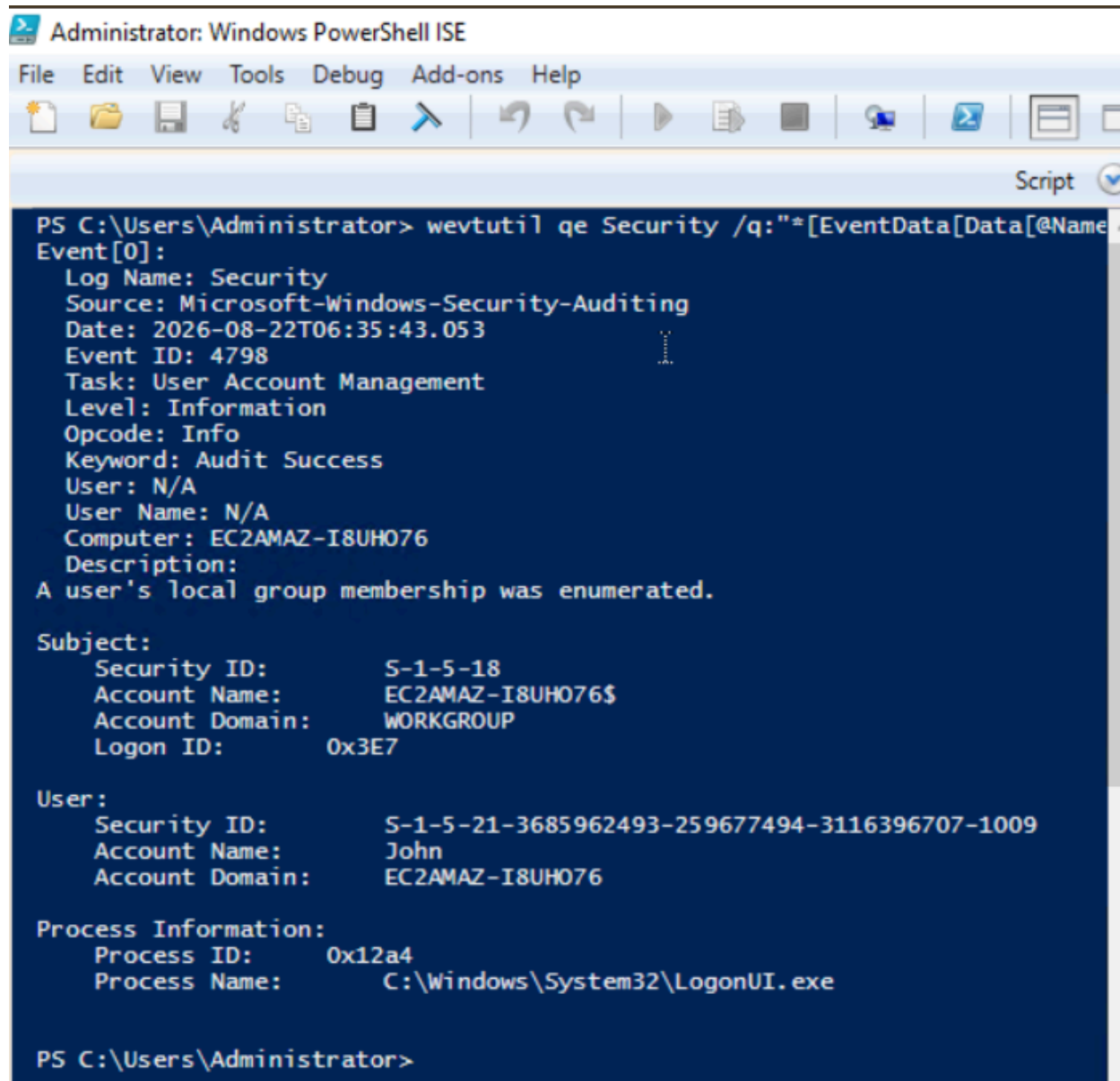
A screenshot of a Windows Command Prompt window titled "Administrator: Command Prompt". The window shows the output of the "systeminfo" command. The text is as follows:

```
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>systeminfo

Host Name:                EC2AMAZ-I8UH076
OS Name:                  Microsoft Windows Server 2016 Datacenter
OS Version:               10.0.14393 N/A Build 14393
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Server
OS Build Type:             Multiprocessor Free
```

John's log in

A screenshot of the Windows PowerShell ISE (Integrated Scripting Environment) window. The title bar reads "Administrator: Windows PowerShell ISE". The menu bar includes "File", "Edit", "View", "Tools", "Debug", "Add-ons", and "Help". The toolbar contains various icons for file operations, editing, and execution. The main console area has a dark blue background and displays the output of a PowerShell command. The command is "PS C:\Users\Administrator> wevtutil qe Security /q:'*[EventData[Data[@Name=Event[0]: Log Name: Security Source: Microsoft-Windows-Security-Auditing Date: 2026-08-22T06:35:43.053 Event ID: 4798 Task: User Account Management Level: Information Opcode: Info Keyword: Audit Success User: N/A User Name: N/A Computer: EC2AMAZ-I8UH076 Description: A user's local group membership was enumerated. Subject: Security ID: S-1-5-18 Account Name: EC2AMAZ-I8UH076\$ Account Domain: WORKGROUP Logon ID: 0x3E7 User: Security ID: S-1-5-21-3685962493-259677494-3116396707-1009 Account Name: John Account Domain: EC2AMAZ-I8UH076 Process Information: Process ID: 0x12a4 Process Name: C:\Windows\System32\LogonUI.exe]']". The prompt "PS C:\Users\Administrator>" is visible at the bottom of the console.

```
Administrator: Windows PowerShell ISE
File Edit View Tools Debug Add-ons Help
Script
PS C:\Users\Administrator> wevtutil qe Security /q:'*[EventData[Data[@Name=Event[0]:
Log Name: Security
Source: Microsoft-Windows-Security-Auditing
Date: 2026-08-22T06:35:43.053
Event ID: 4798
Task: User Account Management
Level: Information
Opcode: Info
Keyword: Audit Success
User: N/A
User Name: N/A
Computer: EC2AMAZ-I8UH076
Description:
A user's local group membership was enumerated.

Subject:
Security ID: S-1-5-18
Account Name: EC2AMAZ-I8UH076$
Account Domain: WORKGROUP
Logon ID: 0x3E7

User:
Security ID: S-1-5-21-3685962493-259677494-3116396707-1009
Account Name: John
Account Domain: EC2AMAZ-I8UH076

Process Information:
Process ID: 0x12a4
Process Name: C:\Windows\System32\LogonUI.exe

PS C:\Users\Administrator>
```

Net user: gets all the users who have logged on

Net user(name): can also be used to get the info o last user

```
Administrator: Windows PowerShell ISE
File Edit View Tools Debug Add-ons Help
[Icons]
Script

The specified local group does not exist.

PS C:\Users\Administrator> net localgroup administartor
net : System error 1376 has occurred.
At line:1 char:1
+ net localgroup administartor
+ ~~~~~
+ CategoryInfo          : NotSpecified: (System error 1376 has occurred) [], RemoteException
+ FullyQualifiedErrorId : NativeCommandError

The specified local group does not exist.

PS C:\Users\Administrator> net localgroup administrator
net : System error 1376 has occurred.
At line:1 char:1
+ net localgroup administrator
+ ~~~~~
+ CategoryInfo          : NotSpecified: (System error 1376 has occurred) [], RemoteException
+ FullyQualifiedErrorId : NativeCommandError

The specified local group does not exist.

PS C:\Users\Administrator> net localgroup administrators
Alias name     administrators
Comment       Administrators have complete and unrestricted access to the
ter/domain

Members

-----
Administrator
Guest
Jenny
ssm-user
The command completed successfully.

PS C:\Users\Administrator>
```

Users with admin privileges apart from admin.

```
Administrator: Windows PowerShell ISE
File Edit View Tools Debug Add-ons Help
Script

PS C:\Users\Administrator> net user jenny
net : The user name could not be found.
At line:1 char:1
+ net user jenny
+ ~~~~~
+ CategoryInfo          : NotSpecified: (The user name could not be fo
:String) [], RemoteException
+ FullyQualifiedErrorId : NativeCommandError

More help is available by typing NET HELPMSG 2221.

PS C:\Users\Administrator> net user jenny
User name                Jenny
Full Name                Jenny
Comment
User's comment
Country/region code      000 (System Default)
Account active           Yes
Account expires          Never

Password last set        3/2/2019 4:52:25 PM
Password expires         Never
Password changeable      3/2/2019 4:52:25 PM
Password required        Yes
User may change password Yes

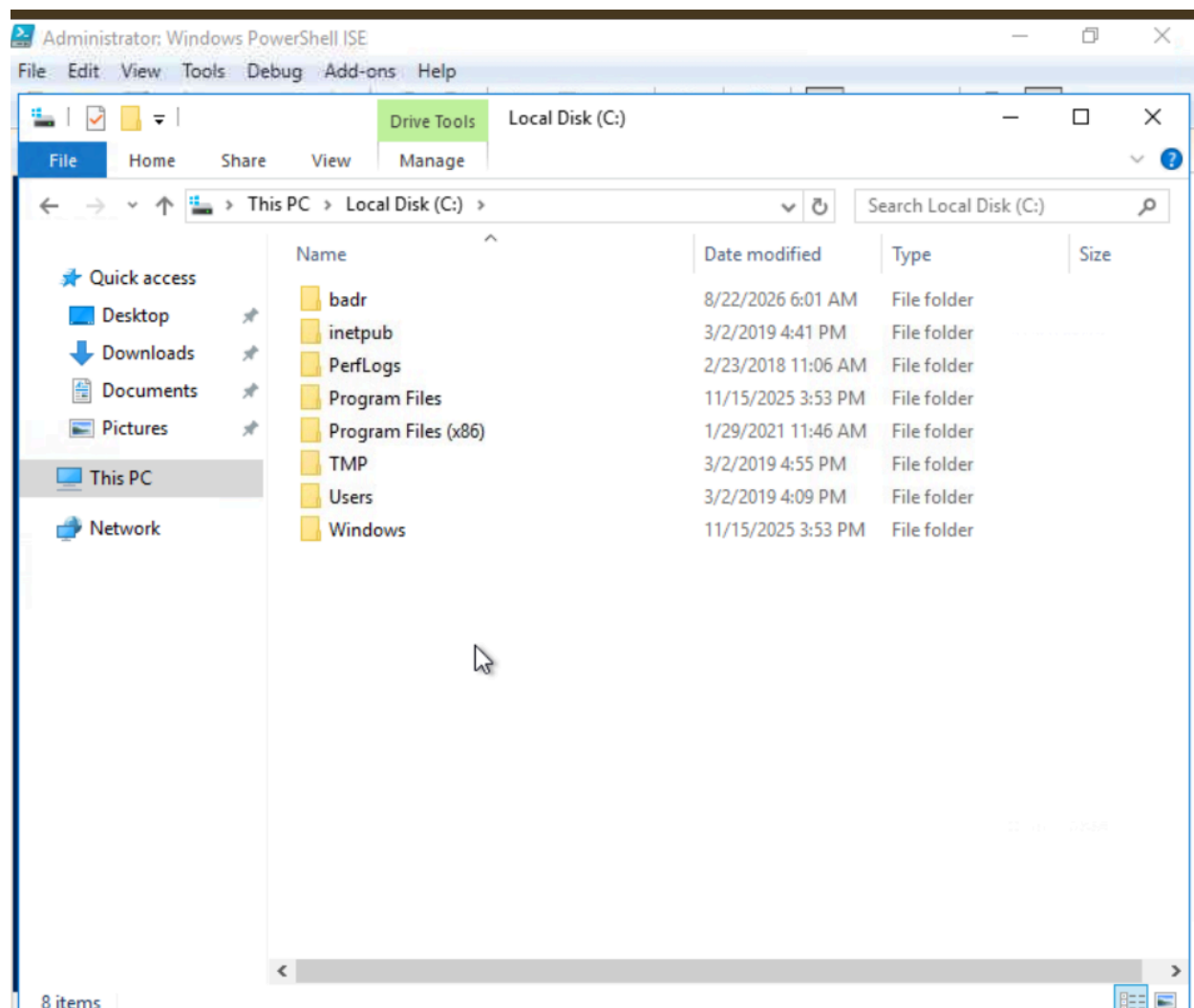
Workstations allowed     All
Logon script
User profile
Home directory
Last logon               Never

Logon hours allowed      All

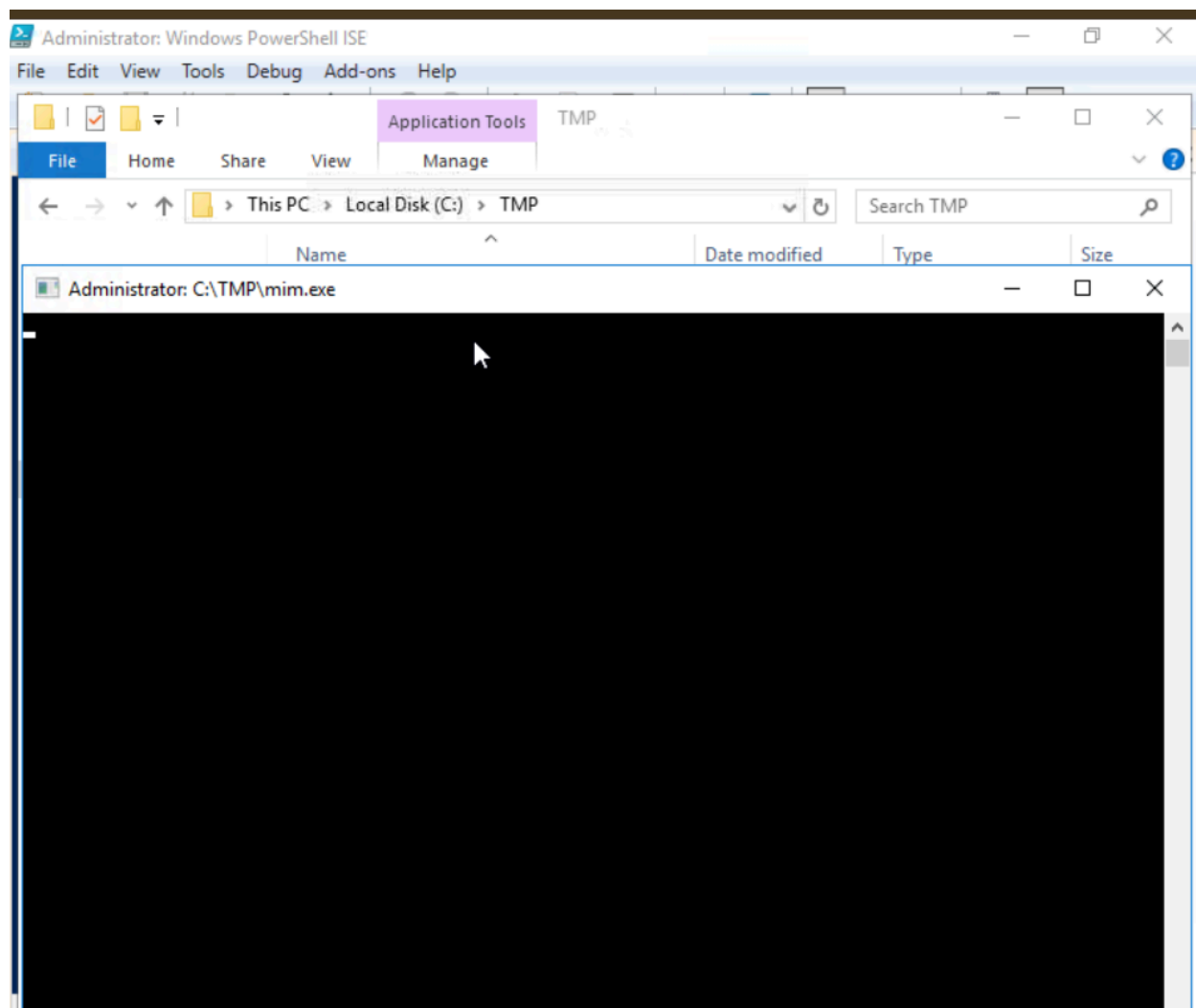
Local Group Memberships  *Administrators
Global Group memberships *None
The command completed successfully.

PS C:\Users\Administrator>
```

Jenny's last logon



The TMP was created at an odd location



Under the tmp file, there was a file with mim, hinting towards imikatz being responsible for windows passwords.